

THE MODEL: THE EFFECTIVE MASS APPROXIMATION AND DONORS IN SI

In this section, we investigate the role of the effective mass approximation in understanding the physics of silicon donors, such as phosphorous. Much of this section is a tweaked version of Shindo and Nara’s paper on “The Effective Mass Equation for the Multi-Valley Semiconductors”, published in the Journal of the Physical Society of Japan, Volume 40, Number 6 on June 1976. We consider their reasoning and help defend or interpret certain conclusions to better understand the role of the various approximations within the effective mass approximation scheme. In this way, we can identify deficiencies within current effective mass theories, and propose strategies for extending current analytical understanding of the silicon donor.

Shallow semiconductor heterostructures in the bulk

The interruption of the crystalline structure of the semiconductor, as with impurities or heterostructures, can serve to capture excess electrons within the semiconductor. We wish to examine the physics of these captured electrons. The Hamiltonian of such a system may be written as

$$\left[\frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 + \hat{U} \right] |\Psi\rangle = E |\Psi\rangle, \quad (1)$$

where $\hat{V}^0$ is the periodic potential of the crystalline lattice, $\hat{U}$ is the potential introduced by a defect in the lattice through an impurity or other means, and E has energy zero at the lowest conduction band minima.

We understand that, in the absence of defect, the wavefunction resulting from the periodic crystal potential will be a Bloch function. The acceptable energies for free electrons occur in discrete bands that may be separated, resulting in disallowed energies. Each Bloch function is denoted by a wavevector and an index indicating membership in one of these bands. The wavefunction of each Bloch state follows the form

$$\langle \mathbf{r} | n\mathbf{k} \rangle = \varphi_{n\mathbf{k}}(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) u_{n\mathbf{k}}(\mathbf{r}), \quad (2)$$

where $u_{n\mathbf{k}}$ may be a complicated function, but one which always has the periodicity of the lattice. The Bloch state is the eigenstate of a nearly free electron travelling through a silicon

lattice. As such, the Schrödinger equation of the pure system may be written as

$$\left[\frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 \right] |n\mathbf{k}\rangle = E_{n\mathbf{k}} |n\mathbf{k}\rangle \quad (3)$$

V^0 is the periodic potential of silicon and $E_{n\mathbf{k}}$ is the energy of a nearly free electron travelling through silicon in the n th conduction band given the wavevector k .

We assume that Bloch functions taken from the first Brillouin zone and from all bands form a complete basis to be used to construct the state of an excess electron bound to an impurity or heterostructure. As such, we represent the complete wavefunction as

$$|\Psi\rangle = \sum_n \int_{BZ} d\mathbf{k} A_n(\mathbf{k}) |n\mathbf{k}\rangle \quad (4)$$

We write the Schrödinger equation in the Bloch basis, letting U be the impurity potential,

$$(E_{n\mathbf{k}} - E) A_n(\mathbf{k}) + \sum_{n'} \int_{BZ} d\mathbf{k}' A_{n'}(\mathbf{k}') \langle n\mathbf{k} | \hat{U} | n'\mathbf{k}' \rangle = 0 \quad (5)$$

We can clearly see the role of the impurity potential in mixing together different bands, as well as causing the state to spread through each band. However, this equation proves difficult to solve directly, since it relies on the interplay between the Fourier components of the impurity potential and the periodic function $u_{n\mathbf{k}}$.

To resolve these difficulties we employ a variational method. In the variational method one calculates $E[\Psi]$ from

$$\langle \Psi | \frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 + \hat{U} - E[\Psi] | \Psi \rangle = 0, \quad (6)$$

where the state Ψ is judiciously chosen, with variational parameters, to solve the problem approximately. Minimization of the energy with respect to the parameters provides the best possible approximate solution to the problem using the state Ψ . Reformulating the Schrödinger equation in light of the variational method yields

$$\sum_n \int_{BZ} d\mathbf{k} |A_n(\mathbf{k})|^2 (E_{n\mathbf{k}} - E) + \sum_{n,n'} \iint_{BZ} d\mathbf{k} d\mathbf{k}' A_n(\mathbf{k})^* A_{n'}(\mathbf{k}') \langle n\mathbf{k} | \hat{U} | n'\mathbf{k}' \rangle = 0. \quad (7)$$

Shallow donor electron states

At this point, we explicitly restrict our concern to “shallow” potentials. For our purposes, the “shallowness” of a potential depends on its effect in mixing and occupying the free

electron states. Namely, the bound state of a shallow potential only significantly occupies the lowest conduction band and occupation within that band is centered steeply around any minima. As such, the bound state possesses non-periodic structure that emerges at a scale much longer than the characteristic wavelength of the Bloch function at the conduction band minima. The shallowness assumptions leads to a series of conclusions:

1. The potential does not cause significant band occupation outside the lowest conduction band, and so terms $n \neq 0$ may be ignored.
2. The band occupation will mostly occur in “valleys” – the k -space energy minima of the lowest conduction band. As such, we write the occupation as $A_0(\mathbf{k}) = \sum_q \alpha_q A^{(q)}(\mathbf{k})$ where the coefficients α_q are determined from group theoretic symmetry concerns and $A^{(q)}(\mathbf{k})$ are sharply peaked functions centered at the q th minimum of the conduction band.

Because we expect the $A^{(q)}(\mathbf{k})$ to be sharply peaked at a particular valley in the conduction band, we can expand $E_n(\mathbf{k})$ in Eq. (7) up to the second order in $\mathbf{k}$ at each valley. Since the valleys are minima, the energy will only contain second order terms, making it analogous to the kinetic energy. This analogy is the source of the effective mass that gives the approximation its name. In general, the minima is anisotropic in $\mathbf{k}$, leading to multiple effective masses.

We wish to see the effect of these assumptions upon Eq. (7). The diagonal part is straight forward, and so we can write

$$\langle \Psi | \frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 - E | \Psi \rangle = \sum_q |\alpha^{(q)}|^2 \int_{BZ} d\mathbf{k} |A^{(q)}(\mathbf{k})|^2 (E^{(q)}(\mathbf{k}) - E) \quad (8)$$

where $E^{(q)}(\mathbf{k})$ is the second order expansion of $E_0(\mathbf{k})$ around the q th valley. Turning our concern to the impurity term in Eq. (7) yields

$$\langle \Psi | \hat{U} | \Psi \rangle = \sum_{q,q'} \alpha_q^* \alpha_{q'} \iint_{BZ} d\mathbf{k} d\mathbf{k}' A^{(q)}(\mathbf{k})^* A^{(q')}(\mathbf{k}') \langle 0\mathbf{k} | \hat{U} | 0\mathbf{k}' \rangle, \quad (9)$$

The matrix element is computed as as

$$\langle 0\mathbf{k} | \hat{U} | 0\mathbf{k}' \rangle = \int d\mathbf{r} U(\mathbf{r}) e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} u_{0\mathbf{k}}^*(\mathbf{r}) u_{0\mathbf{k}'}(\mathbf{r}), \quad (10)$$

so I can write the impurity term as

$$\langle \Psi | \hat{U} | \Psi \rangle = \sum_{q,q'} \alpha_q^* \alpha_{q'} \int d\mathbf{r} U(\mathbf{r}) \iint_{BZ} d\mathbf{k} d\mathbf{k}' A^{(q)}(\mathbf{k})^* e^{-i\mathbf{k}\cdot\mathbf{r}} u_{0\mathbf{k}}^*(\mathbf{r}) A^{(q')}(\mathbf{k}') e^{i\mathbf{k}'\cdot\mathbf{r}} u_{0\mathbf{k}'}(\mathbf{r}) \quad (11)$$

If we wish to examine the long scale physics, it's convenient to introduce a shifted inverse truncated Fourier transform of the occupation function,

$$F^{(q)}(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int_{BZ} d\mathbf{k} A^{(q)}(\mathbf{k}) e^{i(\mathbf{k}-\mathbf{k}_q)\cdot\mathbf{r}} \quad (12)$$

which is often called the “envelope function” since it predicts the long-scale behavior of the wavefunction. With this definition of the envelope function and because $A^{(q)}(\mathbf{k})$ is so steeply peaked about the valley minima, we approximate write the $\mathbf{k}$ -space integrals in the impurity term as

$$F^{(q)}(\mathbf{r}) e^{i\mathbf{k}_q\cdot\mathbf{r}} u_{0\mathbf{k}_q}(\mathbf{r}) \approx \frac{1}{(2\pi)^{3/2}} \int_{BZ} d\mathbf{k} A^{(q)}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} u_{0\mathbf{k}}(\mathbf{r}) \quad (13)$$

With this approximation, we find the impurity term is

$$\langle \Psi | \hat{U} | \Psi \rangle = (2\pi)^3 \sum_{q,q'} \alpha_q^* \alpha_{q'} \int d\mathbf{r} U(\mathbf{r}) F^{(q)}(\mathbf{r})^* e^{-i\mathbf{k}_q\cdot\mathbf{r}} u_{0\mathbf{k}_q}^*(\mathbf{r}) F^{(q')}(\mathbf{r}) e^{i\mathbf{k}_{q'}\cdot\mathbf{r}} u_{0\mathbf{k}_{q'}}(\mathbf{r}) \quad (14)$$

We can also reexpress the diagonal part of the variational Hamiltonian using the envelope function. I introduce an effective kinetic energy operator for each valley $\hat{T}^{(q)}$ such that

$$\hat{T}^{(q)} F^{(q)}(\mathbf{r}) = \frac{1}{(2\pi)^3} \int_{BZ} d\mathbf{k} E^{(q)}(\mathbf{k}) A^{(q)}(\mathbf{k}) e^{i(\mathbf{k}-\mathbf{k}_q)\cdot\mathbf{r}}. \quad (15)$$

As well, it can be shown that

$$\int_{BZ} d\mathbf{k} |A^{(q)}(\mathbf{k})|^2 = \int d\mathbf{r} |F^{(q)}(\mathbf{r})|^2 \quad (16)$$

This allows us to restate equation (8) as

$$\langle \Psi | \frac{\hat{\mathbf{p}}^2}{2m_e} + \hat{V}^0 - E | \Psi \rangle = \sum_q |\alpha^{(q)}|^2 \int d\mathbf{r} F^{(q)}(\mathbf{r})^* \left(\hat{T}^{(q)} - E \right) F^{(q)}(\mathbf{r}) \quad (17)$$

With the state given in terms of the envelope function we can combine Eqs. (14) and (17) as the following variational equation, which may be solved explicitly for the functional expectation energy E ,

$$E = \sum_{q,q'} \alpha_q^* \alpha_{q'} \left(\delta_{qq'} T^{(q)} + U^{(qq')} \right), \quad (18)$$

where

$$T^{(q)} = \int d\mathbf{r} F^{(q)}(\mathbf{r})^* \hat{T}^{(q)} F^{(q)}(\mathbf{r}), \quad (19)$$

$$U^{(qq')} = \int d\mathbf{r} (2\pi)^3 u_{0\mathbf{k}_q}^*(\mathbf{r}) u_{0\mathbf{k}_{q'}}(\mathbf{r}) e^{i(\mathbf{k}_{q'} - \mathbf{k}_q) \cdot \mathbf{r}} F^{(q)}(\mathbf{r})^* F^{(q')}(\mathbf{r}) U(\mathbf{r}), \quad (20)$$

$$\sum_q |\alpha_q|^2 \int d\mathbf{r} |F^{(q)}(\mathbf{r})|^2 = 1. \quad (21)$$

The concern turns now to an adequate choice of functions $F^{(q)}$ so that the variational method may yield adequate results. The choice of $F^{(q)}$ will depend strongly on the band structure of the hosting material. In the absense of information about the host material, we can still represent the envelope function by expanding it in a known basis. As such, we write

$$F^{(q)}(\mathbf{r}) = \sum_i c_i^{(q)} \Phi_i^{(q)}(\mathbf{r}; \Omega_i^{(q)}) \quad (22)$$

We equip our basis with a set of variational parameters Ω to better explore the state space and to minimize the spectrum. If we insert (22) into (18) we are able to write our variational equation as

$$E(\Omega_1, \Omega_2) \sum_{iq, (iq)'} \beta_{iq}^* \beta_{(iq)'} S_{ii'}^{(qq')} = \sum_{iq, (iq)'} \beta_{iq}^* \beta_{(iq)'} \left(T_{ii'}^{(qq')} + U_{ii'}^{(qq')} \right) \quad (23)$$

where S , T , and U are understood as matrices with elements given as

$$S_{ii'}^{(qq')} = \delta_{qq'} \int d\mathbf{r} \Phi_i^{(q)*} \Phi_{i'}^{(q)} \quad (24)$$

$$T_{ii'}^{(qq')} = \delta_{qq'} \int d\mathbf{r} \Phi_i^{(q)*} \hat{T}^q \Phi_{i'}^{(q)} \quad (25)$$

$$U_{ii'}^{(qq')} = \int d\mathbf{r} (2\pi)^3 u_{0\mathbf{k}_q}^* u_{0\mathbf{k}_{q'}} e^{i(\mathbf{k}_{q'} - \mathbf{k}_q) \cdot \mathbf{r}} \Phi_i^{(q)*} \Phi_{i'}^{(q')} U(\mathbf{r}) \quad (26)$$

$$\beta_{iq} = c_i^{(q)} \alpha_q \quad (27)$$

We solve the variational equation above by solving the associated generalized eigenvalue problem,

$$E(\Omega_1, \Omega_2) \mathbf{S} \boldsymbol{\beta} = (\mathbf{T} + \mathbf{U}) \boldsymbol{\beta}, \quad (28)$$

where E and $\boldsymbol{\beta} = (c_i^{(q)} \alpha_q)$ is the generalized eigenvalue and -vector, respectively. The eigenvector may be normalized by satisfying

$$\boldsymbol{\beta}^T \mathbf{S} \boldsymbol{\beta} = 1. \quad (29)$$

From the generalized eigenvalue problem, we acquire information both about the energy and the state associated with a state of variational parameters, and we can write such a state, then, in terms of our basis and our variational parameters as

$$\langle \mathbf{r} | \Psi(\Omega) \rangle \approx \sum_{iq} \beta_{iq} \Phi_i^{(q)}(\mathbf{r}; \Omega_i^{(q)}) e^{i\mathbf{k}_q \cdot \mathbf{r}} (2\pi)^{3/2} u_{\mathbf{k}_q}(\mathbf{r}). \quad (30)$$

The silicon effective mass.

So far the analysis has been somewhat general, applicable to all “shallow” potentials in the semiconductor bulk. The system of most interest in this document is the phosphorous donor in silicon. As such, we want to consider the meaning of “effective mass” in the context of silicon band structure. In silicon, the six equivalent conduction band minima each lie along one of the three coordinate axes, $\hat{k}_x$, $\hat{k}_y$, or $\hat{k}_z$. Additionally, each valley is anisotropic. To the second order in the valley at $\mathbf{k}^{(q)} = k_0 \hat{k}_q$, we find that the conduction band energy goes as

$$E^{(q)}(\mathbf{k}) \approx \frac{\hbar^2}{2m_{\parallel}} (k_q - k_0)^2 + \frac{\hbar^2}{2m_{\perp}} (k_r^2 + k_s^2) \quad (31)$$

The quantities $m_{\parallel}$ and $m_{\perp}$ are defined here from the empirical structure of the silicon conduction band, and they take the role of an effective and anisotropic mass in the kinetic expectation energy. To satisfy Eq. (15) and Eq. (31) we may define the effective kinetic energy operator $\hat{T}^{(q)}$ as

$$\hat{T}^{(q)} = -\frac{\hbar^2}{2m_{\parallel}} \frac{\partial^2}{\partial r_q^2} - \frac{\hbar^2}{2m_{\perp}} \left(\frac{\partial^2}{\partial r_r^2} + \frac{\partial^2}{\partial r_s^2} \right), \quad (32)$$

which will be utilized to find the quantity $T^{(q)}$ as defined in Eq. (19).

We see that the band structure of silicon influences the motion of excess electrons bound to the shallow impurity, emerging as an ellipsoidal anisotropic effective mass. Since an electron occupying a particular bandstructure valley is effectively heavier along the longitudinal axis of the valley, we expect that the individual components of the complete wavefunction corresponding to each valley will show a deformation, leading to an ellipsoidal structure, rather than the spherical structure that might be assumed as the solution to a hydrogenic problem.

Definition of orbital basis states.

We wish to model our system by using a set of basis states that incorporate the hydrogen-like nature of the isolated donor with the anisotropic character of nearly free electrons travelling in the bulk. As such, we consider a set of elliptically scaled Slater or Gaussian orbitals as our variational basis. The states are defined as

$$\begin{aligned} \Phi_{nlm}^{(\pm q)}(\mathbf{r}; \eta \lambda \epsilon) \\ = \mathcal{N}_{nl}(\eta \lambda \epsilon) F\left(\frac{r\vartheta(\chi_q; \lambda)}{\eta}\right) \left[\sum_{j=l}^{n-1} \epsilon_j \left(\frac{r\vartheta(\chi_q; \lambda)}{\eta}\right)^j \right] \hat{P}_{lm}\left(\frac{\lambda \chi_q}{\vartheta(\chi_q; \lambda)}\right) \frac{e^{\pm im\phi_q}}{\sqrt{2\pi}}. \end{aligned} \quad (33)$$

The functions $F(r)$ specify the Slater- or Gaussian-type radial dependence of the underlying prescaled functions, going as

$$F(r) = \exp(-r) \quad \text{or} \quad F(r) = \exp\left(-\frac{r^2}{2}\right) \quad (34)$$

The $\hat{P}_{lm}(\chi)$ denote normalized associated Legendre polynomials. These functions may be more precisely defined as solutions to the following equation

$$\frac{d}{d\chi} \left[(1 - \chi^2) \frac{d}{d\chi} \hat{P}_{lm}(\chi) \right] + \left[l(l+1) - \frac{m^2}{1 - \chi^2} \right] \hat{P}_{lm}(\chi) = 0 \quad (35)$$

with l, m integers such that $0 \leq l \leq n-1$, and $-l \leq m \leq l$. The functions are normalized in the sense that

$$\int_{-1}^{+1} d\chi \hat{P}_{km}(\chi) \hat{P}_{lm}(\chi) = 1. \quad (36)$$

The angular coordinates χ_q and ϕ_q of a basis state are defined by the valley index $q = x, y, z$. Each state has a longitudinal coordinate χ_q associated with the q th valley, namely

$$\chi_x = \frac{x}{r} = \sin(\theta) \cos(\phi), \quad (37)$$

$$\chi_y = \frac{y}{r} = \sin(\theta) \sin(\phi), \quad (38)$$

$$\chi_z = \frac{z}{r} = \cos(\theta), \quad (39)$$

and ϕ_q is defined as the argument of a complex number of unit magnitude, such that

$$e^{\pm im\phi_{q_1}} = \left(\frac{\chi_{q_2} \pm i\chi_{q_3}}{\sqrt{\chi_{q_2}^2 + \chi_{q_3}^2}} \right)^m, \quad (40)$$

q_1	q_2	q_3
x	y	z
y	z	x
z	x	y

The variational parameters are η , λ , and ϵ where η specifies the radial extent of the basis state, and ϵ encodes the node structure. The elliptical anisotropy introduced by the scaling of the valley coordinate of the state expresses itself via the variational parameter λ and the function ϑ which is defined as

$$\vartheta(\chi; \lambda) = \sqrt{1 + (\lambda^2 - 1)\chi^2} \quad (41)$$

and emerges in the $F(r)$ and $\hat{P}_{lm}(\chi)$ as modification of the radial and valley longitudinal coordinates.

Considering the implications of T_d symmetry.

From here, we wish to construct basis states that respect the diamondlike T_d symmetry of our system. In particular, we wish to understand how symmetry constrains our variational parameters. We can denote the variational parameters of a state indexed by valley index q and the state indices nlm as $\Omega_{nlm}^{(q)}$. To ensure our states have T_d symmetry of the system, we would want to assure that our set of basis states remains closed under the operations of the point group. It can be shown that ensuring the variational parameters are such that

$$\Omega_{nlm}^{(q_1)} = \Omega_{nlm}^{(q_2)} = \Omega_{nl(-m)}^{(q_3)}, \quad (42)$$

for all possible values of q_1 , q_2 , and q_3 will satisfy our symmetry requirements. In short, from this we see that the variational parameters depend only on the basis indices $nl|m|$. As such, we can drop the valley index on the variational parameters.